

BOURGES AP, FranceWMO: **072550**Lat: **47.0592N**Lon: **2.3597E**Elev: **529**StdP: **14.42**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	26.2	16.0	12.6	26.6	20.1	15.4	29.2	26.7	51.1	23.5	49.6	6.9	80	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.5	68.7	86.3	67.5	82.7	66.2	70.9	85.2	69.2	82.5	67.7	79.6	7.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.4	75.4	64.8	93.8	73.4	63.5	89.4	72.1	35.0	85.6	33.6	82.7	32.3	79.6	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	17.9	15.6	DB	17.7	96.2	3.9	3.8	14.9	98.9	12.6	101.1	10.4	103.3	7.5	106.1
			WB	17.0	73.9	4.0	1.9	14.1	75.3	11.8	76.4	9.6	77.5	6.7	78.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	39.9	41.2	46.9	51.6	58.4	65.2	68.9	68.4	61.7	55.4	45.8	40.8
	DBStd	12.19	7.22	6.88	6.18	5.95	5.98	6.27	5.81	5.99	5.51	6.55	6.43	7.25
	HDD50	1233	318	252	134	49	6	0	0	0	0	27	155	292
	HDD65	4515	779	668	563	402	218	73	26	28	129	303	576	750
	CDD50	2604	4	4	37	98	266	457	585	572	352	193	29	7
	CDD65	410	0	0	0	1	14	80	146	135	30	4	0	0
	CDH74	4133	0	0	0	30	175	814	1436	1344	301	33	0	0
	CDH80	1442	0	0	0	2	23	272	528	542	72	3	0	0
	WSAvg	7.5	8.5	8.4	8.3	7.8	7.1	6.8	6.9	6.5	6.6	7.1	7.6	8.4
	PrecAvg	28.9	2.4	2.2	2.2	2.2	3.1	2.3	2.3	2.3	2.4	2.6	2.6	2.5
(15) Precipitation	PrecMax	39.7	5.8	5.5	5.5	7.2	6.9	5.0	7.0	6.4	7.0	7.4	5.9	5.8
	PrecMin	19.6	0.2	0.2	0.2	0.3	0.6	0.0	0.3	0.2	0.3	0.2	0.8	0.3
	PrecStd	4.8	1.2	1.1	1.3	1.3	1.5	1.1	1.4	1.5	1.5	1.5	1.1	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	62.8	70.9	77.9	83.0	92.8	94.5	98.5	86.9	78.6	64.5	58.1
		MCWB	52.4	51.9	55.1	61.0	67.0	70.6	69.2	70.2	64.6	63.6	57.5	53.6
	2%	DB	54.5	57.2	65.8	72.4	78.8	86.7	89.8	90.9	81.4	73.2	60.1	55.4
		MCWB	51.5	49.8	53.2	58.0	63.3	68.2	69.0	68.4	64.3	62.4	55.7	52.2
	5%	DB	52.2	53.5	61.5	68.0	75.1	82.5	85.9	85.9	77.1	68.9	57.6	53.3
		MCWB	49.5	48.5	51.5	55.6	62.2	66.8	67.3	67.6	63.0	60.7	54.0	50.8
	10%	DB	50.0	51.1	57.7	63.5	71.0	78.5	82.1	81.4	73.2	65.1	55.1	51.3
		MCWB	47.5	47.4	50.0	53.5	60.1	65.1	65.8	65.9	61.5	59.0	52.2	49.0
	0.4%	WB	54.0	53.7	57.4	62.6	68.6	73.1	72.3	72.7	69.1	65.6	59.0	54.8
		MCDB	55.4	59.4	66.3	75.8	79.8	86.9	87.5	87.0	79.7	73.9	61.3	56.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.9	51.2	54.8	59.3	65.6	70.3	70.4	70.6	66.5	63.4	56.5	52.8
		MCDB	54.1	55.1	62.0	70.1	75.8	83.4	85.1	85.2	76.5	70.3	59.5	54.8
	5%	WB	49.8	49.5	52.9	56.9	63.1	68.2	68.8	68.9	64.6	61.6	54.4	51.1
		MCDB	51.9	52.8	59.3	65.5	72.4	79.2	82.3	82.4	73.4	67.8	57.1	53.0
	10%	WB	47.6	47.7	51.1	54.5	60.9	66.1	67.2	67.1	62.9	59.7	52.3	49.0
		MCDB	49.6	50.9	56.9	61.8	69.2	76.1	79.2	78.8	71.1	64.6	54.9	51.0
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	13.1	16.5	18.0	18.1	18.8	19.5	19.7	18.7	15.7	11.5	10.2
		MCDBR	11.9	18.0	24.0	26.2	25.1	26.0	26.4	27.6	25.9	20.9	14.5	12.1
		MCWBR	9.3	12.3	13.6	13.8	12.6	10.7	9.7	9.9	11.3	11.5	10.2	9.6
		MCDBR	10.9	15.6	19.7	23.5	22.4	23.2	23.4	23.6	20.5	18.7	13.2	11.2
	5% WB	MCWBR	9.1	11.3	12.1	13.0	11.9	10.4	9.2	9.6	10.1	10.9	9.8	9.3
(40) Clear-Sky Solar Irradiance	taub		0.316	0.336	0.364	0.396	0.399	0.410	0.404	0.396	0.364	0.358	0.329	0.313
	taud		2.456	2.399	2.327	2.236	2.277	2.283	2.295	2.334	2.404	2.428	2.456	2.445
	Ebn at Noon		245	263	271	272	275	272	271	269	268	251	237	231
	Edh at Noon		21	28	34	41	41	41	40	37	31	26	21	19
(44) All-Sky Solar Radiation	RadAvg		335	595	993	1397	1642	1883	1820	1594	1232	756	405	284
	RadStd		44	89	120	171	157	176	133	143	93	58	44	48

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A